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Earthquake magnitude

↳ in [earthquake](#) in [The nature of earthquakes](#)

Also known as: *earth tremor*

👤 [Bruce A. Bolt](#) ✓ [Britannica Editors](#) 📅 Sep. 14, 2026 • [History](#)

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Earthquake [magnitude](#) is a measure of the “size,” or [amplitude](#), of the seismic waves generated by an earthquake source and recorded by seismographs. (The types and nature of these waves are described in the section [Seismic waves](#).) Because the size of earthquakes varies enormously, it is necessary for purposes of comparison to compress the range of wave amplitudes measured on seismograms by means of a mathematical device. In 1935 the American seismologist [Charles F. Richter](#) set up a magnitude [scale](#) of earthquakes as the logarithm to base 10 of the maximum [seismic wave](#) amplitude (in thousandths of a millimetre) recorded on a standard [seismograph](#) (the Wood-Anderson torsion pendulum seismograph) at a distance of 100 km (60 miles) from the earthquake epicentre. Reduction of amplitudes observed at various distances to the amplitudes expected at the standard distance of 100 km is made on the basis of [empirical](#) tables. [Richter](#) magnitudes M_L are computed on the assumption that the ratio of the maximum wave amplitudes at two given distances is the same for all earthquakes and is independent of azimuth.



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Richter first applied his magnitude scale to shallow-focus earthquakes recorded within 600 km of the epicentre in the southern

[California region](#). Later,

additional empirical tables were set up, whereby observations made at distant stations and on seismographs other than the standard type could be used. Empirical tables were extended to cover earthquakes of all significant focal depths and to enable independent magnitude estimates to be made from body- and surface-wave observations. A current form of the Richter scale is shown in the table.

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Richter scale of earthquake magnitude			
magnitude level	category	effects	earthquakes per year
less than 1.0 to 2.9	micro	generally not felt by people, though recorded on local instruments	more than 100,000
3.0–3.9	minor	felt by many people; no damage	12,000–100,000
4.0–4.9	light	felt by all; minor breakage of objects	2,000–12,000
5.0–5.9	moderate	some damage to weak structures	200–2,000
6.0–6.9	strong	moderate damage in populated areas	20–200
7.0–7.9	major	serious damage over large areas; loss of life	3–20
8.0 and higher	great	severe destruction and loss of life over large areas	fewer than 3

At the present time a number of different magnitude scales are used by scientists and engineers as a measure of the relative size of an earthquake. The *P*-wave magnitude (M_b), for one, is defined in terms of the [amplitude](#) of the *P* wave recorded on a standard seismograph. Similarly, the surface-wave magnitude (M_s) is defined in terms of the [logarithm](#) of the maximum amplitude of ground motion for surface waves with a wave period of 20 seconds.

As defined, an earthquake magnitude scale has no lower or upper limit. Sensitive seismographs can record earthquakes with magnitudes of negative value and have recorded magnitudes up to about 9.0. (The [1906 San Francisco earthquake](#), for example, had a Richter magnitude of 8.25.)

A scientific weakness is that there is no direct mechanical basis for magnitude as defined above. Rather, it is an empirical [parameter analogous](#) to stellar magnitude assessed by astronomers. In modern practice a more soundly based mechanical measure of earthquake size is used—namely, the seismic moment (M_0). Such a parameter is related to the angular leverage of the forces that produce the slip on the causative [fault](#). It can be calculated both from recorded seismic waves and from field measurements of the size of the fault rupture. Consequently, seismic moment provides a more uniform scale of earthquake size based on classical mechanics. This measure allows a more scientific magnitude to be used called [moment magnitude](#) (M_w). It is proportional to the logarithm of the seismic moment; [values](#) do not differ greatly from M_s values for moderate earthquakes. Given the above definitions, the great [Alaska earthquake of 1964](#), with a Richter magnitude (M_L) of 8.3, also had the values $M_s = 8.4$, $M_0 = 820 \times 10^{27}$ dyne centimetres, and $M_w = 9.2$.

Earthquake energy

Energy in an earthquake passing a particular surface site can be calculated directly from the recordings of seismic ground motion, given, for example, as ground velocity. Such recordings indicate an energy rate of 10^5 [watts](#) per square metre (9,300 watts per square foot) near a moderate-size earthquake source. The total power output of a rupturing fault in a shallow earthquake is on the order of 10^{14} watts, compared with the 10^5 watts generated in rocket motors.

The surface-wave magnitude M_s has also been connected with the surface energy E_s of an earthquake by [empirical](#) formulas. These give $E_s = 6.3 \times 10^{11}$ and 1.4×10^{25} [ergs](#) for earthquakes of $M_s = 0$ and 8.9, respectively. A unit increase in M_s corresponds to approximately a 32-fold increase in energy. Negative magnitudes M_s

correspond to the smallest instrumentally recorded earthquakes, a magnitude of 1.5 to the smallest felt earthquakes, and one of 3.0 to any shock felt at a distance of up to 20 km (12 miles). Earthquakes of magnitude 5.0 cause light damage near the [epicentre](#); those of 6.0 are destructive over a restricted area; and those of 7.5 are at the lower limit of major earthquakes.

The total annual [energy](#) released in all earthquakes is about 10^{25} ergs, corresponding to a rate of work between 10 million and 100 million kilowatts. This is approximately one one-thousandth the annual amount of heat escaping from the [Earth](#)'s interior. Ninety percent of the total seismic energy comes from earthquakes of magnitude 7.0 and higher—that is, those whose energy is on the order of 10^{23} ergs or more.

Frequency

There also are empirical relations for the frequencies of earthquakes of various magnitudes. Suppose N to be the average number of shocks per year for which the magnitude lies in a range about M_s . Then

$$\log_{10} N = a - bM_s$$

fits the data well both globally and for particular regions; for example, for shallow earthquakes worldwide, $a = 6.7$ and $b = 0.9$ when $M_s > 6.0$. The frequency for larger earthquakes therefore increases by a factor of about 10 when the magnitude is diminished by one unit. The increase in frequency with reduction in M_s falls short, however, of matching the decrease in the energy E . Thus, larger earthquakes are overwhelmingly responsible for most of the total seismic energy release. The number of earthquakes per year with $M_b > 4.0$ reaches 50,000.

Occurrence of earthquakes

Tectonic associations

Global [seismicity](#) patterns had no strong theoretical explanation until the [dynamic](#) model called [plate tectonics](#) was developed during the late 1960s. This theory holds that the Earth's upper shell, or [lithosphere](#), consists of nearly a dozen large, quasi-stable slabs called plates. The thickness of each of these plates is roughly 80 km (50 miles). The plates move horizontally relative to neighbouring plates at a rate of 1 to 10 cm (0.4 to 4 inches) per year over a shell of lesser strength called the [asthenosphere](#). At the plate edges where there is contact between adjoining plates, boundary tectonic forces operate on the rocks, causing physical and chemical changes in them. New lithosphere is created at [oceanic ridges](#) by the upwelling and cooling of [magma](#) from the Earth's [mantle](#). The horizontally moving plates are believed to be absorbed at the [ocean](#) trenches, where a [subduction](#) process carries the [lithosphere](#) downward into the Earth's interior. The total amount of lithospheric material destroyed at these [subduction zones](#) equals that generated at the ridges.

Seismological evidence (such as the location of major earthquake belts) is everywhere in agreement with this tectonic model.

Earthquake sources are concentrated along the [oceanic ridges](#), which correspond to [divergent plate boundaries](#). At the subduction zones, which are associated with [convergent plate boundaries](#), intermediate- and deep-focus earthquakes mark the location of the upper part of a dipping lithosphere slab. The focal mechanisms indicate that the stresses are aligned with the dip of the lithosphere underneath the [adjacent continent](#) or [island arc](#).

Some earthquakes associated with oceanic ridges are confined to strike-slip faults, called transform faults, that offset the ridge crests. The majority of the earthquakes occurring along such horizontal shear faults are characterized by slip motions. Also in agreement with the plate tectonics theory is the high seismicity encountered along the edges of plates where they slide past each other. Plate boundaries of this kind, sometimes called fracture zones, include the [San Andreas Fault](#) in [California](#) and the North Anatolian fault system in [Turkey](#). Such plate boundaries are the site of interplate earthquakes of shallow focus.

The low seismicity within plates is consistent with the plate tectonic description. Small to large earthquakes do occur in limited regions well within the boundaries of plates; however, such intraplate seismic events can be explained by tectonic mechanisms other than plate boundary motions and their associated phenomena.

Shallow, intermediate, and deep foci

Most parts of the world experience at least occasional shallow earthquakes—those that originate within 60 km (40 miles) of the Earth's outer surface. In fact, the great majority of earthquake foci are shallow. It should be noted, however, that the geographic distribution of smaller earthquakes is less completely determined

than more severe quakes, partly because the availability of relevant [data](#) is dependent on the distribution of observatories.



Of the total energy released in earthquakes, 12 percent comes from intermediate earthquakes—that is, quakes with a focal depth ranging from about 60 to 300 km. About 3 percent of total energy comes from deeper earthquakes. The frequency of occurrence falls off rapidly with increasing focal depth in the intermediate range. Below intermediate depth the distribution is fairly uniform until the greatest focal depths, of about 700 km (430 miles), are approached.

The deeper-focus earthquakes commonly occur in patterns called [Benioff zones](#) that dip into the Earth, indicating the presence of a [subducting](#) slab. Dip angles of these slabs average about 45°, with some shallower and others nearly vertical. Benioff zones coincide with tectonically active [island arcs](#) such as [Japan](#), [Vanuatu](#), [Tonga](#), and the [Aleutians](#), and they are normally but not always associated with deep [ocean trenches](#) such as those along the South American [Andes](#). Exceptions to this rule include [Romania](#) and the [Hindu Kush mountain](#) system. In most Benioff zones, intermediate- and deep-earthquake foci lie in a narrow layer, although recent precise hypocentral locations in Japan and elsewhere show two distinct parallel bands of foci 20 km apart.



Aftershocks, foreshocks, and swarms

Usually, a major or even [moderate](#) earthquake of shallow focus is followed by many lesser-size earthquakes close to the original source [region](#). This is to be expected if the [fault](#) rupture producing a major earthquake does not relieve all the accumulated strain energy at once. In fact, this dislocation is liable to cause an increase in the stress and strain at a number of places in the vicinity of the focal region, bringing crustal rocks at certain points close to the stress at which fracture occurs. In some cases an earthquake may be followed by 1,000 or more [aftershocks](#) a day.

Sometimes a large earthquake is followed by a similar one along the same fault source within an hour or perhaps a day. An extreme case of this is multiple earthquakes. In most instances, however, the first principal earthquake of a series is much more severe than the [aftershocks](#). In general, the number of aftershocks per day decreases with time. The [aftershock](#) frequency is roughly inversely proportional to the time since the occurrence of the largest earthquake of the series.



Most major earthquakes occur without detectable warning, but some principal earthquakes are preceded by foreshocks. In another common pattern, large numbers of small earthquakes may occur in a region for months without a major earthquake. In the Matsushiro region of [Japan](#), for instance, there occurred between August 1965 and August 1967 a series of hundreds of thousands of earthquakes, some sufficiently strong (up to Richter magnitude 5) to cause property damage but no casualties. The maximum frequency was 6,780 small earthquakes on April 17, 1966. Such series of

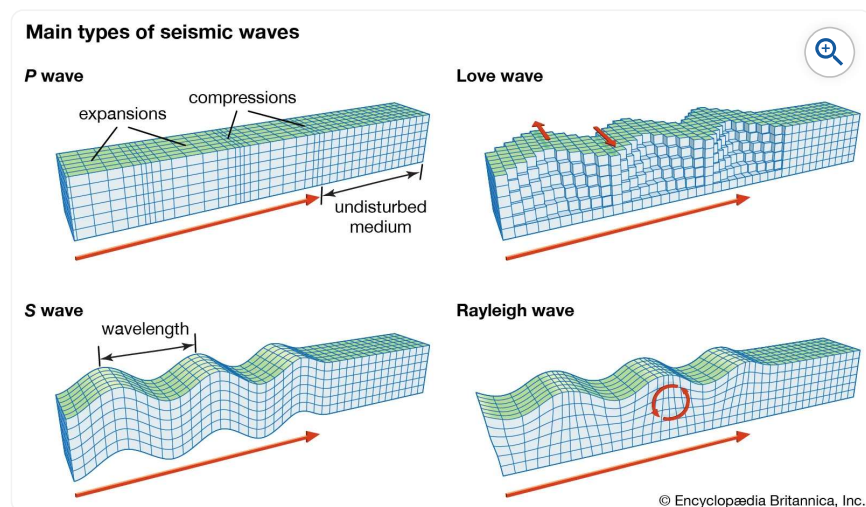
earthquakes are called earthquake swarms. Earthquakes associated with [volcanic activity](#) often occur in swarms, though swarms also have been observed in many nonvolcanic regions.



The study of earthquakes

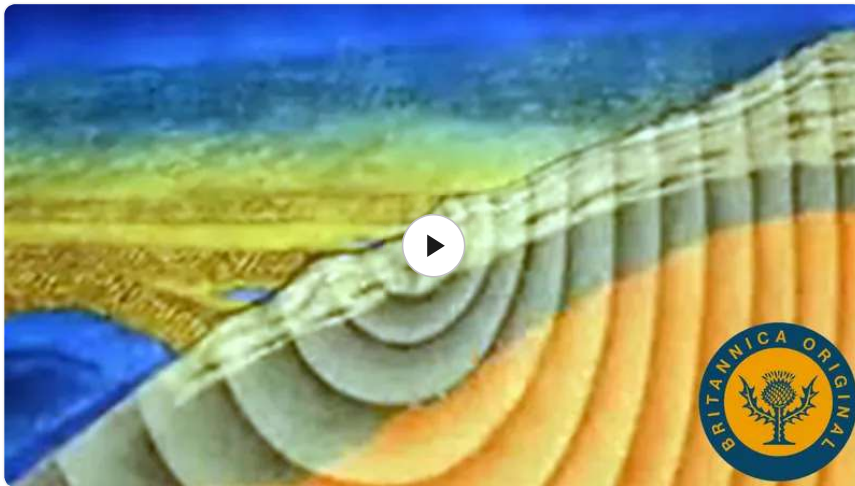
Seismic waves

Principal types of seismic waves



seismic wave: main types Diagram showing the main types of seismic waves: *P* (primary), *S* (secondary), Love, and Rayleigh. [... \(more\)](#)

Seismic waves generated by an earthquake source are commonly classified into three main types. The first two, the *P* (or primary) and *S* (or secondary) waves, [propagate](#) within the body of the [Earth](#), while the third, consisting of Love and Rayleigh waves, [propagates](#) along its surface. The existence of these types of seismic waves was mathematically predicted during the 19th century, and modern comparisons show that there is a close correspondence between such theoretical calculations and actual measurements of the seismic waves.



Learn about P waves, S waves, Love waves, and Rayleigh waves generated by an earthquake Seismic waves travel in different patterns and at different speeds through the Earth. The most damaging are the surface types—the Love waves and the Ri...(more)

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The [P seismic waves](#) travel as elastic motions at the highest speeds. They are [longitudinal waves](#) that can be transmitted by both [solid](#) and [liquid](#) materials in the Earth's interior. With *P* waves, the particles of the medium vibrate in a manner similar to sound waves—the transmitting media is alternately compressed and expanded. The slower type of [body wave](#), the [S wave](#), travels only through solid material. With *S* waves, the particle motion is [transverse](#) to the direction of travel and involves a [shearing](#) of the transmitting rock.



Because of their greater speed, *P* waves are the first to reach any point on the Earth's surface. The first *P*-wave onset starts from the spot where an earthquake originates. This point, usually at some depth within the Earth, is called the [focus](#), or hypocentre. The point at the surface immediately above the focus is known as the [epicentre](#).

[Love](#) and [Rayleigh waves](#) are guided by the free surface of the Earth. They follow along after the *P* and *S* waves have passed through the body of the planet. Both Love and Rayleigh waves involve horizontal particle motion, but only the latter type has vertical ground [displacements](#). As Love and Rayleigh waves travel, they disperse into long wave trains, and, at substantial distances from the source in alluvial basins, they cause much of the shaking felt during earthquakes.

